

Analyzing Maryland Insurance Appeals and Grievances: A Data Dashboard Project

Kenny Nguyen

DATA 205 – 33155

Professor Perine

Maryland Insurance Administration



Introduction

Maryland Insurance Administration (MIA)

Data: 2021 – 2024 Appeals and Grievances
report on Maryland.gov

Adverse Decision: When an insurance claim is
rejected

Grievance: Appeal of the denied claim

Complaints: If grievances get denied and still
unsatisfied, formal complaints directly to MIA

Goals and Tools



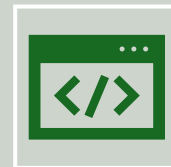
Analyze trends in adverse decisions and grievances over time and to identify key patterns.



Identify patterns and trends in complaints outcomes and decision



Why it matters? \$194 billion to \$220 billion in medical debt



Adobe Acrobat, Excel, R-Studio, PowerBI

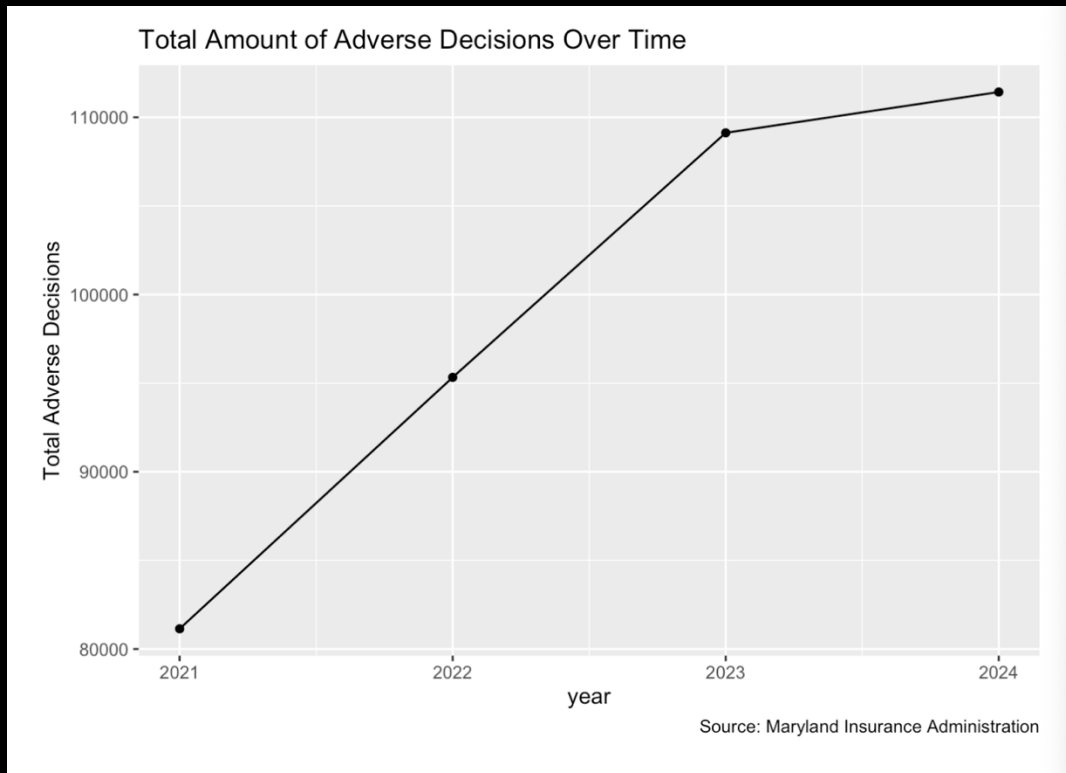
Data Ingestion/ Cleaning

COMPANY NAME	ADVERSE DECISIONS		A. INPATIENT HOSPITAL SERVICES		B. EMERGENCY ROOM SERVICES	
	COMPANY TOTAL	% OF ALL COMPANIES	NUMBER	% TOTAL	NUMBER	% TOTAL
4 Ever Life Insurance Company	1	0.0%	1	100.0%	0	0.0%
Aetna Dental, Inc.	693	0.9%	0	0.0%	0	0.0%
Aetna Health Inc. (a Pennsylvania corporation)	115	0.1%	39	33.9%	0	0.0%
Aetna Life Insurance Company	188	0.2%	69	36.7%	0	0.0%
Ameritas Life Insurance Corp.	285	0.4%	0	0.0%	0	0.0%
CareFirst BlueChoice, Inc.	21,128	26.0%	12	0.1%	0	0.0%
CareFirst of Maryland, Inc.	7,057	8.7%	2	0.0%	1	0.0%
CIGNA Dental Health of Maryland, Inc.	47	0.1%	0	0.0%	0	0.0%
CIGNA Health and Life Insurance Company	17,081	21.1%	284	1.7%	0	0.0%
Connecticut General Life Insurance Company	7	0.0%	0	0.0%	0	0.0%
Delta Dental Insurance Company	8	0.0%	0	0.0%	0	0.0%
Delta Dental of Pennsylvania	55	0.1%	0	0.0%	0	0.0%
Dentega Insurance Company	9	0.0%	0	0.0%	0	0.0%
Dental Network, Inc. The	8	0.0%	0	0.0%	0	0.0%
Dominion Dental Services, Inc.	1,111	1.4%	0	0.0%	0	0.0%
Golden Rule Insurance Company	12	0.0%	0	0.0%	0	0.0%
Group Dental Service of Maryland, Inc.	1,922	2.4%	0	0.0%	0	0.0%
Group Hospitalization and Medical Services, Inc.	5,515	6.8%	6	0.1%	0	0.0%
Guardian Life Insurance Company of America	1,263	1.6%	0	0.0%	0	0.0%
Johns Hopkins HealthCare LLC	90	0.1%	16	17.8%	0	0.0%
Kaiser Foundation Health Plan-Mid-Atlantic States, Inc.	812	1.0%	33	4.1%	0	0.0%
Kaiser Permanente Insurance Company	61	0.1%	24	39.3%	0	0.0%
Lincoln Life and Annuity Company of New York	1	0.0%	0	0.0%	0	0.0%
Lincoln National Life Insurance Company	117	0.1%	0	0.0%	0	0.0%
MAMSI Life and Health Insurance Company	916	1.1%	7	0.8%	0	0.0%
Metropolitan Life Insurance Company	668	0.8%	0	0.0%	0	0.0%
National Health Life Insurance Company	5	0.0%	2	40.0%	0	0.0%
Optimum Choice, Inc.	2,376	2.9%	101	4.3%	1	0.0%

[illegible]

How Has The Number of Adverse Decisions Changed Over Time?

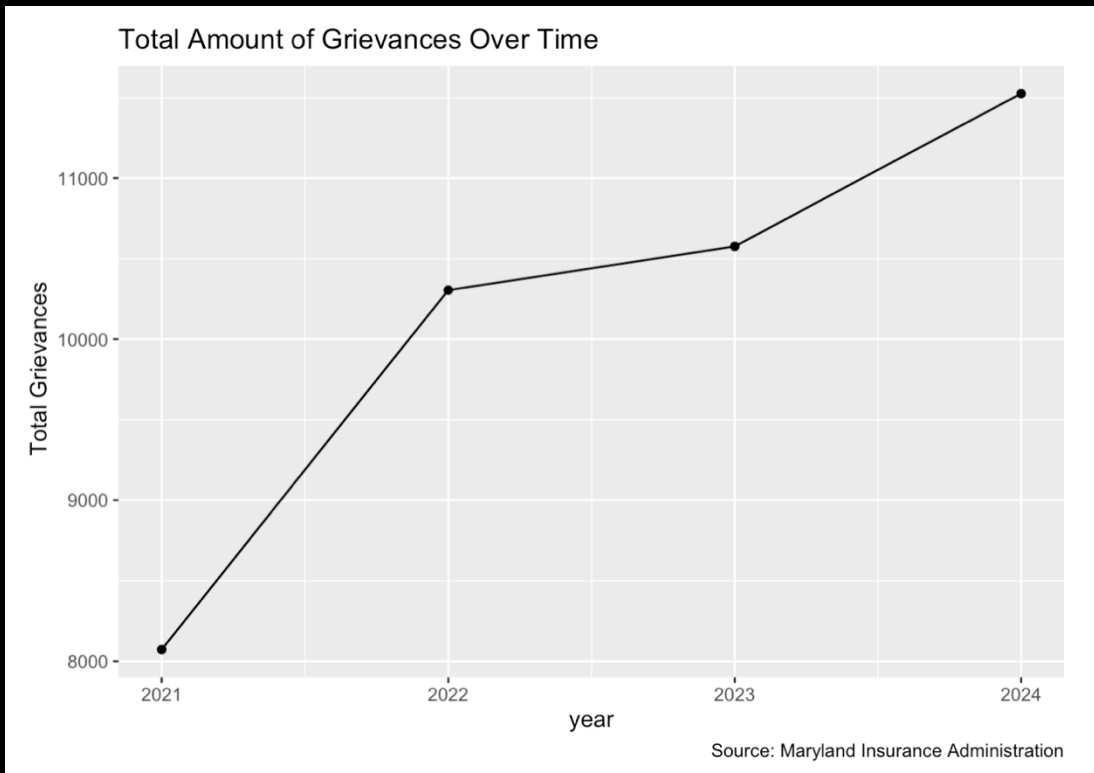
- Adverse decisions increased steadily from 81,143 in 2021 to 111,426 in 2024, with the largest growth occurring between 2021 and 2022 (17.48%)
- Growth slowed significantly in 2024 (2.11%), suggesting a potential stabilization in the volume of adverse decisions



```
adverse_decisions_company_total  
Min.      : 81143  
1st Qu.: 91781  
Median :102225  
Mean      : 99255  
3rd Qu.:109699  
Max.      :111426
```

How Has The Number of Grievances Changed Over Time?

- Total grievances grew 43% from 8,073 in 2021 to 11,526 in 2024, indicating consumers are increasingly challenging insurance decisions.



```
grievances_filed_company_total  
Min.      : 8073  
1st Qu.: 9747  
Median : 10441  
Mean     : 10120  
3rd Qu.: 10814  
Max.     : 11526
```

DASHBOARD DEMO!





Conclusion

- Increase in both adverse decisions and grievances
 - Rate of growth stabilized
 - Pharmacy services impact
- High success rates for complaints

Acknowledgments

Professor Saidi

Professor Alraee

Professor Perine

Professor Maliha

Brent Matthewson

Louis Buttler

Riley Williams

Classmates

References

Maryland Insurance Administration. (2025). *2024 report on the Health Care Appeals and Grievance Law*.
<https://insurance.maryland.gov/Consumer/Appeals%20and%20Grievances%20Reports/Maryland-Insurance-Administration-Annual-Report-FY-2025.pdf>

Maryland Insurance Administration. (2024, December 1). *2023 report on the Health Care Appeals and Grievance Law*.
<https://insurance.maryland.gov/Consumer/Appeals%20and%20Grievances%20Reports/2023-Report-on-the-Health-Care-Appeals-and-Grievance-Law.pdf>

Maryland Insurance Administration. (2023). *2022 report on the Health Care Appeals and Grievance Law*.
<https://insurance.maryland.gov/Consumer/Appeals%20and%20Grievances%20Reports/FY-2023-Maryland-Insurance-Administration-Annual-Report.pdf>

Maryland Insurance Administration. (2021). *2021 report on the Health Care Appeals and Grievance Law*.
<https://insurance.maryland.gov/Consumer/Appeals%20and%20Grievances%20Reports/FY-2021-Maryland-Insurance-Administration-Annual-Report.pdf>



**Thank You,
Questions?**